Modelling Moon Phases · Pupil Resources

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Arrival Questions (In your book)

1. In the model, what does the fixed torch stand for?

2. What does the white polystyrene ball stand for?

3. What does the observer in the middle of the circle stand for?

4. Can you ask an adult to help?

Starter Question

Why does holding the ball slightly above head height at Position 5 make the model more accurate?

A. It models the 5-degree tilt of the Moon's real orbit, allowing sunlight to reach the Full Moon without an eclipse.

B. It makes the ball turn into gas.

C. It changes the torch battery.

Method Strip

1. Check the torch is fixed pointing across the mat from the SUN marker.

2. Hold the white ball at Position 1 (towards the torch) and observe which side is dark. STOP AND CHECK.

3. Step through Positions 2, 3, 4, and 5 (holding Pos 5 slightly higher so your shadow does not block the

light).

4. Step through Positions 6, 7, and 8 to complete the 28-day orbit.

5. Shade and label all 8 phase circles and write your model evaluation in your book.

Exit Ticket

In our model, what did the torch, the observer, and the ball represent?

Write the four main phases in order starting from New Moon.

State one strength and one limitation of the ball-and-torch model.